Hyung Jiwon

Email | GitHub | LinkedIn

Seoul, South Korea

Education

- **Ewha Womans University**

  B.S. in Data Science — Seoul, South Korea

Research Interests

Natural Language Processing and Large Language Models; Retrieval-Augmented Generation (RAG); Self-Improving Agents; LLM Alignment; AI Security.

Selected Projects

- **Intent-Aware Search Result Filtering** — *Nov. 2026 - Present*

  NLP, Neural Re-ranking, Chrome Extension | In Progress

  - Designing a Chrome extension that scores query-document relevance to filter search results while allowing users to restore filtered content.
  - Planning comparisons of keyword, embedding, and neural re-ranking approaches, prioritizing relevant-document retention and inference latency.

- **Intersection Risk Assessment for Traffic Safety Policy** — *Aug. 2026*

  Python, pandas, SciPy, statsmodels, scikit-learn **[Code]**

  - Led the integration of six public data sources across 72 intersections and designed a negative binomial and Empirical Bayes framework to assess excess crash risk.
  - Achieved **ROC-AUC 0.841** on a 2024 temporal holdout, supporting risk-based prioritization for traffic safety policy.

- **Privacy Proxy: NLP-Based Privacy Protection for LLMs**

  Python, Named Entity Recognition, Regular Expressions **[Code]**

  - Collaborated on Korean NER model development and developed post-processing to integrate NER and rule-based privacy detections, resolving overlapping spans based on detection evidence.
  - Used a unified schema of **27 entity types** to assess combined risk from location, age, and purchase information and implemented risk-based retention, generalization, and tokenization policies.

Skills

- **Programming:** Python, R, C
- **ML Frameworks:** PyTorch, Hugging Face
- **Libraries & Tools:** NumPy, pandas, SciPy, statsmodels, scikit-learn, Jupyter Notebook

Activities

- **BOAZ (AI Club)** — *Jul. 2026 - Present*

  27th Staff

- **KISIA AI Security Technology Development Program** — *Jul. 2026 - Nov. 2026*

  Personal Information Protection Track

  - Training and project work on AI-based personal information leakage detection and protection models.

- **Ewha Student Council** — *Mar. 2026 - Jun. 2026*

  Public Relations Team Member